

MOUSTAPHA BA

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LinkedIn — GitHub

Summary

AI and software engineering student driven by a personal mission to use technology to close critical gaps in healthcare access. Motivated by the loss of my grandmother from a misdiagnosis due to limited technological resources, I have already built and deployed high-impact AI systems, including AI nurse agents powering mobile healthcare units that expand care access in underserved regions. With hands-on experience across AI, data-driven systems, and scalable software engineering, I am focused on designing reliable, ethical, and real-world AI solutions—and scaling them to reach the people who need them most.

Education

Virginia Tech — B.S., Computational Modeling and Data Analytics Expected 05/2028
GPA: 3.8

Experience

BaLabs — Co-Founder 08/2025 – Present
Architect and deploy AI-driven automation systems for businesses, focusing on workflow optimization, customer engagement, and scalable system design. Lead end-to-end development from solution architecture to deployment, delivering measurable operational improvements.

NelCura Healthcare — Leader of AI Operations 08/2024 – Present
Lead development of AI systems for an AI-powered mobile healthcare initiative expanding access to care in underserved regions. Designed intelligent agents to automate patient intake, clinical triage, and diagnosis support, emphasizing scalability, data integrity, and safe AI-to-physician handoff.

Microsoft — AI Engineering Intern 06/2024 – 09/2024
Built AI prototypes using Python, OpenAI APIs, and Azure services while collaborating in agile teams to solve real-world problems under production constraints.

Nivaro — Software Engineer Intern 01/2024 – 08/2024
Contributed to front-end engineering and product development at an AI startup, improving functionality, user experience, and software reliability through iterative development.

Projects and Certifications

- **AI Nurse Agents – NelCura:** Built and deployed AI nurse agents enabling real-time symptom analysis, triage, and physician handoff within AI-powered mobile healthcare units, expanding access to quality care in underserved communities across Africa.
- **MIT Pokerbots Competition:** Developed an autonomous Texas Hold'em poker bot implementing probabilistic reasoning and strategic decision-making under uncertainty; achieved **2nd place out of 90+ teams**.
- **IBA AI – Personalized Autism Care Chatbot:** Built a personalized AI chatbot using retrieval-augmented generation (RAG) pipelines with OpenAI APIs and custom datasets to deliver accurate, context-aware caregiver support.
- **StudentCoders of VA:** Founded and led a nonprofit initiative providing computer science education to underserved students across Virginia, designing curriculum and scaling online programs to improve technical literacy and long-term access to technology.
- Microsoft Azure AI Engineer Certificate